

The Third Workshop on Multiple Input Modalities and Sensations for VR/AR Interactions (MIMSVAI)

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ABSTRACT

Rapid technological advances are expanding the practical applicability of virtual reality and/or augmented reality (VR/AR); however, the degree to which new users are able to interact with these technologies is limited by current input modalities. Gaining an intuitive grasp of VR/AR applications requires that users be immersed in the virtual environment, which in turn depends on the integration of multiple realistic sensory feedback mechanisms. This workshop will bring together researchers from the fields of UbiComp and VR/AR to explore alternative input modalities and sensory feedback systems to facilitate the design of coherent and engaging VR/AR experiences comparable to those in the real world.

CCS CONCEPTS

• Human-centered computing → Ubiquitous and mobile computing systems and tools.

KEYWORDS

Virtual reality; Augmented reality; Multiple sensory modalities; Multi-sensations; Mixed reality; Realism; User friendliness.

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1 OBJECTIVE

With the advancements in Virtual Reality and/or Augmented Reality (VR/AR) technology, an increasing number of VR and AR applications are being developed and gaining immense popularity among new users [7]. To enhance user experience and interaction with these systems, it is essential to provide multiple user input modalities. Additionally, integrating multiple realistic sensations cohesively can significantly improve the level of realism, thus creating a truly immersive VR/AR experience [2]. However, the lack of robust and intuitive interaction interfaces and realistic sensations can impede users' experience and prevent the widespread acceptance of VR/AR interactions across different application areas. Therefore, many researchers are dedicating their efforts to explore how to achieve a higher level of coherence between different input modalities and sensations, aiming to provide more engaging VR/AR experiences.

Immersion in a cohesive virtual reality and/or augmented reality (VR/AR) environment depends on multiple realistic sensations, whereas the ability to interact with the system depends on an unobtrusive interface and intuitive input modalities. Rapid technological advances have greatly improved the user experience; however, VR/AR system developers are striving to further enhance the realism of VR/AR applications by incorporating multiple sensory inputs, including visual systems of greater sophistication [3], haptic feedback [10], auditory cues [4], and olfactory sensations [6]. Developers are also seeking to employ a diversity of input modalities, such as eye gaze [5], voice [8], and gesture [11] to enable seamless interactions with the virtual/augmented environment. Nonetheless, VR/AR developers are struggling to provide users with an experience that is as realistic and immersive as those experienced in the real world.

The Multiple Input Modalities and Sensations for VR/AR Interactions (MIMSVAI 2023) workshop will be a one-day face-to-face gathering of researchers in the fields of ubiquitous computing (UbiComp) and VR/AR. Note that attendees planning to attend the workshop in Cancun Mexico on October 8, 2023 will be expected to register for the UbiComp/ISWC 2023 main conference. Our initial efforts will be aimed at developing research momentum in the UbiComp community by calling for papers on the design of input modalities and sensory feedback mechanisms aimed at enhancing the immersive quality of the VR/AR experience and facilitating

computer-user interactions. We arrange sessions for each accepted paper to be presented in the workshop. At the end of each session, we will assign session chairs to mediate the discussion to distill motivated ideas and findings to the participants. Besides the regular sessions, we will invite one expert for VR/AR interactions to give talks in the morning. We hope the keynote presentation can bring in more interesting perspectives regarding VR/AR-related researches. In this year, we further incorporate discussion sessions in the afternoon to engage workshop attendees in productive brainstorming and discussion to generate more potential challenges or ideas to enable more immersive VR/AR experiences. Finally, we will deliver the closing remarks and give the best paper award to the best paper presented at the MIMSVAI 2023 workshop to acknowledge and encourage excellence in research. The awardee will be presented with a certificate at the workshop.

2 RELEVANT AND TIMELY

UbiComp refers to the integration of computing in the lives and physical environment of users. It is important to note however that UbiComp is fundamentally different from virtual reality [9]. UbiComp involves the use of wearable computing devices in the real world, whereas virtual reality involves immersing the user in a computer-generated virtual environment. Augmented reality (AR) falls between these two extremes by superimposing a virtual world on the real world to enhance our perception of the surrounding environment. VR replaces the real world with a virtual environment, whereas AR enhances our perception of the real world.

Note that technological advancements and the cross-pollinating of ideas have blurred the lines between UbiComp, VR, and AR. The UbiComp components that allow user interactions in the physical world (via mobile and wearable devices) can also be used to allow interaction in virtual worlds (via VR/AR devices). It is important to consider that despite fundamental differences in usage, all these devices depend on human-computer interactions. This workshop will provide an opportunity for researchers in the fields of UbiComp and VR/AR to share their ideas pertaining to the design of input modalities and sensory feedback mechanisms aimed at facilitating VR/AR interactions.

3 TOPICS

Our primary objective is to facilitate conversations and collaborations among experts in the fields of UbiComp and VR/AR. The main topics will include the following: (1) Enhancing user-computer interactions via improved input modalities; (2) Enhancing the realism of virtual worlds via sensory feedback; (3) Developing new devices and infrastructure; (4) Identifying new application scenarios; and (5) Expanding design processes, platforms, and tools for immersive VR/AR interactions. We will be petitioning submissions related to these topics but will also be open to other related areas of inquiry. Workshop discussions will also focus on the five topics mentioned above.

- **Improving input modalities.** Input systems are meant to detect user actions and interpret their intentions as they interact with a virtual world. Advances in this area will require improving or abandoning existing input modes or combining them with new ones.

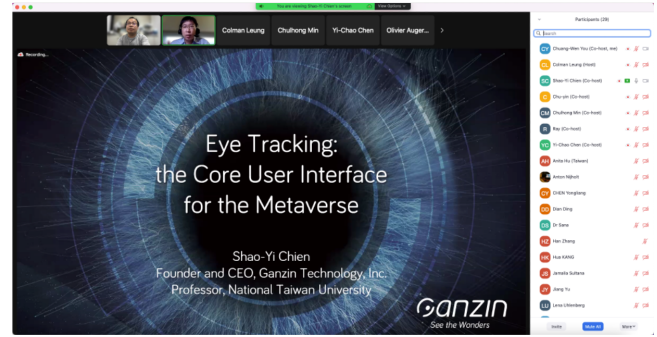


Figure 1: Keynote speech presented in MIMSVAI 2022 (Virtual Event).

- **Enhancing the realism of virtual worlds via sensory feedback.** The aim in creating a virtual world is to generate an immersive experience in which the user feels that they are truly interacting with virtual objects in a manner that mimics their experiences in the real world. Advances in this area will require developing new techniques and building upon existing ones.
- **Developing new devices and infrastructures.** Researchers will need to develop new and innovative approaches to VR/AR devices and infrastructure that go beyond the computational constraints of current first-generation devices.
- **Identifying new application scenarios.** VR/AR technology has already found its way into gaming, healthcare, and education; however, these immersive technologies are still in their infancy. It is expected that 5 to 10 years will be required for these technologies to approach their full potential. In the coming years, we can expect to see the application of VR/AR technology in ways that we cannot currently imagine. Researchers must adopt a visionary approach to exploring and expanding the application of these emerging technologies.
- **Expanding design process, platform/tools, or libraries.** Creating new VR/AR applications will involve predicting user requirements and identifying how those needs can be met. It will necessitate significant modifications to current platforms/tools (such as Unity) and libraries (such as Android ARCore or Apple ARKit), as well as the invention of completely new techniques.

4 WORKSHOP STRUCTURE

Following on the success of MIMSVAI 22, the upcoming MIMSVAI 2023 workshop will be a one-day workshop intended for in-person participation. The inclusion of interactive activities, keynote speeches, and presentations by participants (Table 1) will ensure that this year's event is as engaging and informative as last year's (Figs. 1 and 2). Workshop sessions will be scheduled for the presentation of all accepted papers, at the end of which discussions and brainstorming activities will be held under the mediation of a session chair. We will also be inviting experts in the field of VR/AR interaction to give keynote presentations in the morning. During the closing remarks, we will be awarding a certificate to the authors of the best paper presented at the MIMSVAI 2023 workshop.

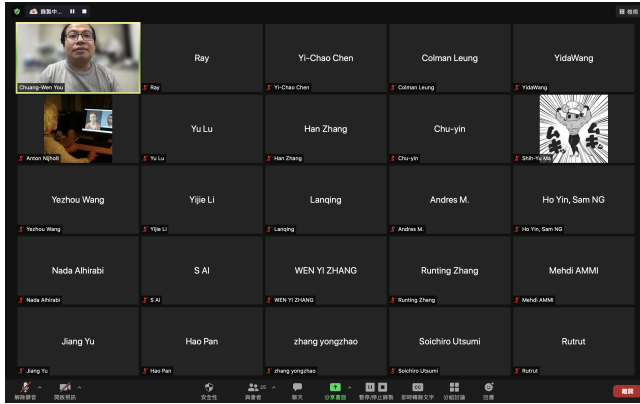


Figure 2: Photo of online attendees from MIMSVAI 2022 (Virtual Event).

Table 1: Workshop Schedule

Workshop Schedule	
9:00-9:10 am	Opening
9:10am-10:20am	Keynote Talk I: 50 + 20min
10:20am-10:30am	Coffee Break
10:30am-11:10am	Paper Session I: 2 × (15 + 5)min
11:10am-11:20am	Coffee Break
11:20am-12:00pm	Paper Session II: 2 × (15 + 5)min
12:00pm-2:00pm	Lunch Break
2:00pm-2:40pm	Paper Session III: 2 × (15 + 5)min
2:40pm-2:50pm	Coffee Break
2:50pm-3:50pm	Discussion Session I: 60min
3:50pm-4:00pm	Coffee Break
4:00pm-5:00pm	Discussion Session II: 60min
5:00pm-5:10pm	Closing & Best Paper Award Ceremony

Submissions should not exceed 5 pages in length (4 pages for content and 1 page for references), following the ACM sigconf template [1]. Accepted papers will be published in the UbiComp/ISWC Adjunct Proceedings included in the ACM Digital Library. If the paper is accepted, it will be published in the UbiComp/ISWC Adjunct Proceedings, which is included in the ACM Digital Library. A Best Paper Award will be presented to recognize and encourage outstanding research at the end of MIMSVAI 2023 workshop. The recipient of the award will receive a certificate during the workshop.

The call for papers will be sent to a variety of research communities related to UbiComp, HCI, VR/AR applications, haptic and tactile interfaces, and multi-sensory perception. Submitted papers will be reviewed by at least two reviewers from the Program Committee and organization staff. Papers will be evaluated based on their novelty, quality, and relevance. We expect to accept four papers for presentation at the workshop. Due to the theme of the workshop and the rapid growth of the VR/AR community, we anticipate a large number of submissions, which should allow us to organize a high-quality event.

5 EXPECTED OUTCOMES

We expect that at least 20 participants will be attending the workshop, including individuals who submit papers as well as researchers with an interest in VR/AR technology. It is hoped that this workshop will deepen our understanding of state-of-the-art methods as well as new developments in the future.

ACKNOWLEDGMENTS

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A APPENDIX: CALL FOR PAPERS

The increasing popularity of VR/AR technology has led to the emergence of numerous VR/AR applications that appeal to new users. To enhance the level of perceived realism, alternative input modalities and multiple realistic sensations need to be developed and integrated coherently for users to interact with these technologies. Although these developments have the potential to create more immersive VR/AR experiences, the lack of robust and intuitive interaction interfaces and realistic sensations limits the level of users' acceptance in various application areas. This workshop aims to address the challenges and opportunities of designing a higher coherence between different input modalities and sensations to offer more engaging VR/AR experiences. It provides a platform for

researchers from both UbiComp and VR/AR fields to collaborate and explore alternative input modalities and sensations for VR/AR interactions.

Papers may include some of the topics, but are not limited to:

- 2D/3D and volumetric display and projection technology
- Immersive analytics and visualization
- Modeling and simulation
- Multimodal capturing and reconstruction
- Scene description and management issues
- Storytelling
- Tracking and sensing
- Audio interfaces, sound rendering, spatialized audio, auditory perception, and psychoacoustics
- Embodied agents, virtual humans, and self-avatars
- Haptic and tactile interfaces, wearable haptics, passive haptics, pseudo haptics, and other touch-based user interfaces
- Mediated and diminished reality
- Multimodal input and output
- Multisensory rendering, registration, and synchronization
- Perception and cognition
- Presence, body ownership, and agency
- Teleoperation and telepresence
- 3D user interaction
- 3D user interface metaphors
- Collaborative interactions
- Ethical issues
- Human factors and ergonomics
- Input devices
- Locomotion and navigation
- Multimodal/cross-modal interaction and perception
- Non-fatiguing 3D user interfaces
- Non-visual interfaces (such as olfactory)
- Touch, tangible, and gesture interfaces
- Usage research, evaluation methods, and empirical studies
- Domain-specific VR/AR applications.
- Design process, platform/tools, or libraries used to incorporate input modalities or sensations for VR/AR interactions or studying their use.

Min-Chun Hu, National Tsing Hua University

Wei Sun, Samsung Research America

Yun-Jui Lee, Croma New Media

Important Dates

Paper submission: June 2, 2023

Paper notification: June 30, 2023

Camera-ready: July 31, 2023

Workshop: October 8, 2023

Best Paper Award

The Best Paper Award is given to the best paper presented at the MIMSAI 2023 workshop, to acknowledge and encourage excellence in research. The awardee will be presented with a certificate at the workshop.

Workshop Organizer

Chuang-Wen You, National Tsing Hua University

Yi-Chao Chen, Shanghai Jiao Tong University

Liwei Chan, National Yang Ming Chiao Tung University